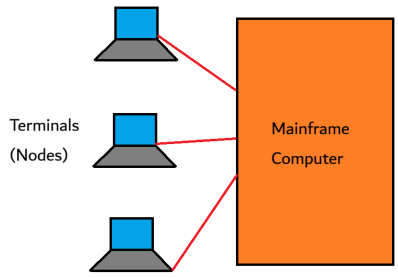
Introduction to Distributed Systems

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1 Centralized Computing

Early form of computing was centralized , Also called Uniprocessor computing.



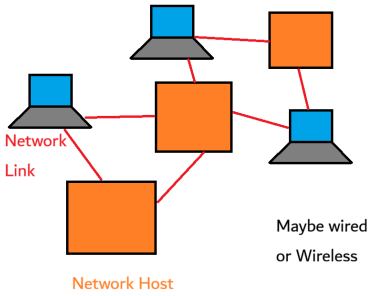
2 What is distributed system ?

Collection of independent computers interconnected via network collaborating on tasks

2.1 Another declaration

Collection of autonomous computing elements that appears to users as Single coherent system

Each node is autonomous and have it's own notion of time , there is no global clock.



3 Coherent system

The nodes in system as whole operates the same.

No matters where or when or how users and system interaction takes a place.

4 Overlay Network

Logical network which each node communicates with set of others node called **neighbors**.

P2P Systems is good example for overlay networks

4.1 Types of overlay networks

- Structured : Well-defined set of neighbors
- Unstructured : Randomly selected neighbors

5 Computers in Distributed Systems

- Workstations : Used by End-users to perform computing
- Servers : Provides services and resources
- PADs (Personal Assistance Devices)

5.1 Examples of Distributed Systems

- Internet
- ATM Machines
- Intranets / Workgroups

5.2 Types of Distribution Systems

- High performance distributed systems (Parallel - Grid - Cluster - Cloud)
- Distributed Information Systems (Middleware-EAI-TPM)
- Distributed systems for pervasive computing (Ubitiqouis - Mobile - Sensor networks)

6 Characteristics of Distributed Systems

1. Heterogeneity are mostly hidden
2. Users interact consistently and uniformly with system
3. Easy to expand or scale
4. It's normally continuous avaliable

7 Main goals of Distributed Systems

- Sharing of resources and services
- Distribution Transparency
- Openness
- Scalability

7.1 Scalability

Be Able to add more users and devices to system
And be able to separate users and devices geographically.

7.1.1 Scaling Techniques

- Hiding communication latencies
- Distribution which splitting it into smaller parts and subsequently spreading
- Replication components across a distributed system

8 Challenges of Distributed Systems

- Heterogeneity (2 Kinds of OS)
- Latency
- Synchronization
- DoS Attacks
- Partial Failure